# Planning Investigations and Representing Data

Name: ________________________ Date: _____________ Year 4 Science — Science Inquiry Skills

ACARA v9.0: AC9S4I02 / AC9S4I04

### Learning Intention

I am learning to plan a fair investigation and represent my results using a simple table or column graph.

### Success Criteria

- I can identify the elements of a fair test.
- I can record results in a table.
- I can represent data using a simple column graph.

### Plan It, Test It, Show It

A good investigation: (1) changes only ONE variable, (2) keeps everything else the same, (3) records results in a table, and (4) can show results in a column graph to spot patterns.

### Worked Example

Testing how far 3 toy cars roll down the same ramp:
Car A: 40cm, Car B: 65cm, Car C: 50cm
This can be shown as a column graph with 3 bars.

### Part A — Warm Up

Circle the best way to show and compare number results visually: **a paragraph** **a column graph** **a drawing**

True or False: A fair test should change 2 or more variables at once. ______

Name one tool used to make a formal measurement (e.g. length, mass, temperature). ______

### Part B — Core Skills

Record in a table: Plant A grew 4cm, Plant B grew 7cm, Plant C grew 5cm.

________________________________

Why is a column graph easier to compare than a list of numbers?

________________________________

Name 2 things to keep the same testing paper towel absorbency.
1. ____________ 2. ____________

Why compare your results with a classmate's?

________________________________

### Word Bank

fair test • variable • table • column graph • data • measurement • pattern • conclusion • compare

### Working Space — Draw a Simple Column Graph for 3 Made-Up Results

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### Part C — Challenge: The Paper Plane Investigation

> *A Year 4 class tests 3 differently shaped paper planes to see which flies furthest. Each plane is thrown 3 times by the same student, from the same spot, using the same amount of force, and the distances are recorded in a table.*

Q8. Identify the one variable being changed on purpose in this investigation.

Q9. Name 2 things the class correctly kept the same to make this a fair test.

Q10. Plane B flew the furthest in all 3 throws. A student wants to show this clearly to the class in one picture. Suggest the best way to represent this data, and explain why it is better than just reading out the numbers.

### Answer Key — Part A

**A1:** a column graph.

**A2:** False — a fair test should only change one variable at a time so the results are reliable.

**A3:** Ruler, thermometer, scales, or measuring jug (accept any reasonable measuring tool).

### Answer Key — Part B

**B1:** A table showing Plant A: 4cm, Plant B: 7cm, Plant C: 5cm (teacher to check formatting).

**B2:** A column graph shows the differences visually with bars of different heights, making it quicker and easier to compare sizes at a glance than reading through a list of numbers.

**B3:** The amount of water used and the size of the paper towel (accept any 2 reasonable controlled factors).

**B4:** Comparing results helps check if findings are similar (making the investigation more reliable) or different (which might mean something went wrong or needs further investigation).

### Answer Key — Part C

**C8:** The shape of the paper plane.

**C9:** Same student throwing, same starting spot, same amount of force, same number of throws (accept any 2 reasonable factors kept the same).

**C10:** A column graph — it would show all 3 planes' distances as bars side by side, making it instantly clear which plane flew furthest without needing to read and compare individual numbers.

### I Can Checklist

☐ I can identify elements of a fair test.

☐ I can record results in a table.

☐ I can represent data in a column graph.